
Inverse Contextual Bandits: Learning How Behavior Evolves over Time

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Abstract

Understanding a decision-maker’s priorities by observing their behavior is critical for transparency and accountability in decision processes—such as in healthcare. Though conventional approaches to policy learning almost invariably assume stationarity in behavior, this is hardly true in practice: Medical practice is constantly evolving as clinical professionals fine-tune their knowledge over time. For instance, as the medical community’s understanding of organ transplantations has progressed over the years, a pertinent question is: How have actual organ allocation policies been evolving? To give an answer, we desire a policy learning method that provides *interpretable* representations of decision-making, in particular capturing an agent’s *non-stationary* knowledge of the world, as well as operating in an *offline* manner. First, we model the evolving behavior of decision-makers in terms of contextual bandits, and formalize the problem of *Inverse Contextual Bandits* (ICB). Second, we propose two concrete algorithms as solutions, learning parametric and nonparametric representations of an agent’s behavior. Finally, using both real and simulated data for liver transplantations, we illustrate the applicability and explainability of our method, as well as benchmarking and validating its accuracy.

1. Introduction

Modeling decision-making policies is a central concern in computational and behavioral science, with key applications in healthcare [1], economics [2], and cognition [3]. The business of *policy learning* is to determine an agent’s decision-making policy given observations of their behavior.

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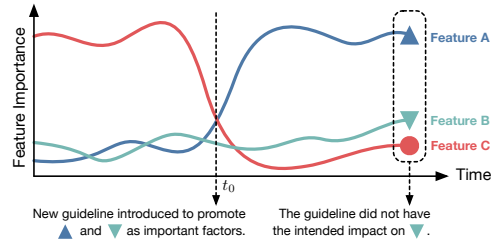


Figure 1. *Evaluating Impact of Policies via ICB.* Policy-makers introduce a new clinical guideline to promote Features A and B as main factors of consideration in making a certain decision, while previously, clinicians relied on Feature C instead. The new guideline succeeds at establishing Feature A as such but not Feature B. Using, ICB we can infer how the importance of these features have evolved over time and observe quantitatively that Feature B indeed did not gain importance despite the original intentions. Now, the policy-makers can revise their guideline accordingly.

Typically, the objective is either to replicate the behavior of some demonstrator (cf. “imitation learning”) [4, 5], or to match their performance on the basis of some reward function (cf. “apprenticeship learning”) [6, 7]. However, equally important is the pursuit of “descriptive modeling” [8, 9]—that is, of learning interpretable parameterizations of behavior for auditing, quantifying, and understanding decision-making policies. For instance, recent work has studied representing behaviors in terms of rules [10], goals [11], intentions [12], preferences [13], subjective dynamics [14], as well as counterfactual outcomes [15].

Evolving Behaviors In this work, we ask a novel descriptive question: *How has the observed behavior changed over time?* While conventional approaches to policy learning almost invariably assume that decision-making agents are stationary, this is rarely the case in practice: In many settings, behaviors evolve constantly as decision-makers learn more about their environment and adjust their policies accordingly. In fact, disseminating new knowledge from medical research into actual clinical practice is itself a major endeavor in healthcare [16, 17, 18]. This research question is new: While capturing “variation in practice” in observed data has been studied in the context of demonstrations containing mixed policies [19, 20], multiple tasks [21, 22], and subgroup-/patient-specific preferences [8], little work has attempted to capture variation in practice *over time* as an agent’s knowledge of the world evolves.

Example (Organ Allocations) As our core application, consider organ allocation practices for liver transplantations: In the medical community, our understanding of organ transplantations has changed numerous times over past decades [23, 24, 25]. Thus an immediate question lies in *how* actual organ allocation practices have changed over the years. Having a data-driven, quantitative, and—importantly—interpretable description of how practices have evolved is crucial: It would enable policy-makers to objectively evaluate if the policies they introduced have had the intended impact on practice; this would in turn play a substantial role in designing better policies going forward (see Figure 1) [8].

To tackle questions of this form, we desire a policy learning method that satisfies three key desiderata: It should provide an (i.) *interpretable* description of observed behavior. It should be able to capture an agent’s (ii.) *non-stationary* knowledge of the world. It should operate in an (iii.) *offline* manner—since online experimentation is impossible in high-stakes environments such as healthcare.

Inverse Contextual Bandits To accomplish this, we first identify the organ allocation problem as a *contextual bandits* problem [26, 27, 28]: Given each arriving instance of patient and organ features (i.e., the context), an agent makes an allocation decision (i.e. the action), whence some measure of feedback is perceived (i.e., internal reward). Crucially, the environment (i.e., precisely, its reward dynamics) is unknown to the agent and must be *actively learned*, so the agent maintains beliefs about their environment (i.e., internal knowledge). Not only must an agent select actions that “exploit” the knowledge they have, but they must also select actions that “explore” the environment to update their knowledge. Thus the behavior of a learning agent is naturally modeled as a generalized bandit strategy—that is, how to take actions based on their knowledge, and how to update their knowledge based on the outcomes of their actions.

Now, the *forward* contextual bandits problem asks the (normative) question: Given an unknown environment, what is an effective bandit strategy that minimizes some notion of regret? By contrast, our focus here is instead on the opposite direction—that is, in formalizing and solving the problem of *Inverse Contextual Bandits* (“ICB”): We ask the (descriptive) question: Given demonstrated behavior from a decision-making agent, how has the agent’s knowledge been evolving over time? Precisely, we wish to learn interpretable representations of generalized bandit strategies from the observable context-action pairs generated by those strategies—regardless of whether they are effective.

Contributions Our contributions are three-fold. In the sequel, we first formalize the ICB problem, identifying it with the data-driven objective of inferring an agent’s internal reward function along with their internal trajectory of beliefs about the environment (Section 2). Second, we propose

two learning algorithms, imposing different specifications regarding the agent’s behavioral strategy: The first parameterizes the agent’s knowledge in terms of Bayesian updates, whereas the second makes the milder specification that the agent’s behavior evolves smoothly over time (Section 3). Third, through both simulated and real-world data for liver transplantations, we illustrate how ICB can be applied as an investigative device for recovering and explaining the evolution of organ allocation practices over the years, as well as validating the accuracy of our algorithms (Section 5).

2. Inverse Contextual Bandits

Preliminaries Consider a *decision-making problem* of the form $\mathbb{D} := (X, A, \mathcal{R}, \mathcal{T})$, where X indicates the state space, A the action space, $\mathcal{R} \in \Delta(\mathbb{R})^{X \times A}$ the reward dynamics, and $\mathcal{T} \in \Delta(X)^{X \times A}$ the transition dynamics. At each time $t \in \mathbb{Z}_+$, the decision-making agent is presented with some state $x_t \in X$ and decides to take an action $a_t \in A$, whence an immediate reward $r_t \sim \mathcal{R}(x_t, a_t)$ is received, and a subsequent state $x_{t+1} \sim \mathcal{T}(x_t, a_t)$ is presented. Let the *decision-making policy* employed by the agent be denoted $\pi \in \Delta(A)^X$, actions are sampled according to $a_t \sim \pi(x_t)$.

In the forward direction, given reward dynamics $\mathcal{R}$ and transition dynamics $\mathcal{T}$, the *reinforcement learning* problem (“RL”) deals with determining the optimal policy that maximizes some notion of expected cumulative rewards [29]: $\pi_{\mathcal{R}, \mathcal{T}}^* := \operatorname{argmax}_{\pi \in \Delta(A)^X} \mathbb{E}_{\pi, \mathcal{R}, \mathcal{T}} \sum_t r_t$. In the opposite direction, given some observed behavior policy π_b and transition dynamics $\mathcal{T}$, the *inverse reinforcement learning* problem (“IRL”) deals with determining the reward dynamics $\mathcal{R}$ with respect to which π_b appears optimal. For instance, the classic max-margin approach seeks [30]: $\mathcal{R}^* := \operatorname{argmin}_{\mathcal{R}} (\max_{\pi} \mathbb{E}_{\pi, \mathcal{R}, \mathcal{T}} \sum_t r_t - \mathbb{E}_{\pi_b, \mathcal{R}, \mathcal{T}} \sum_t r_t)$.

Problems with No Dynamics Access In conventional RL (and IRL), the decision-maker has unrestricted access to environment dynamics—either explicitly (i.e., $\mathcal{R}, \mathcal{T}$ are simply *known* and used in computing the optimal policy), or implicitly (i.e., the agent may *interact* freely with the environment during training). In contrast, in our setting the agent has no such luxuries—not only are the dynamics not known to the agent, but neither do they enjoy a distinct sandboxed training phase prior to live deployment. Without such access, the agent must consider both the information they may gain when taking actions (cf. “exploration”) and also the expected rewards due to those actions (cf. “exploitation”). Note that this property results in much more difficult learning problems—in both the forward and inverse directions (see Appendix B for a more detailed discussion).

Consider environments parameterized by *environment parameters* $\rho \in P$, such that the reward dynamics of an environment are given by $\mathcal{R}_\rho$, and the transition dynamics by $\mathcal{T}_\rho$.

Let ρ_{env} denote the true environment parameter, such that the actual rewards received by an agent are distributed as $r_t \sim \mathcal{R}_{\rho_{\text{env}}}(x_t, a_t)$, and the actual states encountered by the agent are distributed as $x_{t+1} \sim \mathcal{T}_{\rho_{\text{env}}}(x_t, a_t)$. Since ρ_{env} is unknown by the agent, they take actions on the basis of their beliefs about it; these beliefs are described by probability distributions $\mathcal{P}_\beta \in \Delta(P)$ over environment parameters, and parameterized by *belief parameters* $\beta \in B$. For each time t , let β_t capture the agent’s belief at the beginning of that step.

Each time step, the agent first samples an environment parameter $\rho_t \sim \mathcal{P}_{\beta_t}$ according to their belief β_t , then takes an action according to $\pi_{\mathcal{R}_{\rho_t}, \mathcal{T}_{\rho_t}}^*$, which is the optimal policy under the sampled environment parameter. This ensures that each action is taken with probability proportional to the probability with which the agent believes it to be optimal. Essentially, the agent’s policy π_t at time t is induced by their belief β_t such that $\pi_t(x)[a] = \pi_{\beta_t}(x)[a] := \mathbb{E}_{\rho \sim \mathcal{P}_{\beta_t}}[\pi_{\mathcal{R}_\rho, \mathcal{T}_\rho}^*(x)[a]]$. After receiving a reward r_t , the agent updates their current belief parameter according to a (possibly-stochastic) *belief-update function* $f \in \Delta(B)^{B \times X \times A \times \mathbb{R}}$, such that $\beta_{t+1} \sim f(\beta_t, x_t, a_t, r_t)$. Together with the initial belief parameter β_1 , the agent’s belief at time t is a (possibly-stochastic) function of the history $h_t = \{x_{1:t-1}, a_{1:t-1}, r_{1:t-1}\}$ defined recursively via f .

2.1. Contextual Bandits Setting

In this work, we consider state transitions that occur independently of past states and actions. Due to this property, decisions can be made greedily without consideration of what future states may be, and yields a contextual bandits problem [26, 27, 28]. Note that this captures the organ allocation problem well: Distributions of newly arriving organs are largely independent of prior allocation decisions; in fact, transplantation and allocation policies are often modeled in bandit-like settings [31, 32]. Formally:

Definition 1 (Contextual Bandits) Consider a decision-making problem $\mathbb{D} := (X, A, \mathcal{R}, \mathcal{T})$, where $\mathcal{R}, \mathcal{T}$ are unknown to the agent. Let $\mathcal{T}(x, a) = \mathcal{T}'$ for some $\mathcal{T}' \in \Delta(X)$, for all $x \in X, a \in A$, such that policies are greedy: $\text{supp}(\pi_t^*(x_t)) = \arg\max_{a_t \in A} \bar{\mathcal{R}}_{\rho_t}(x_t, a_t)$, where $\bar{\mathcal{R}}_\rho(x, a) := \mathbb{E}_{r \sim \mathcal{R}_\rho(x, a)}[r]$ indicates the mean reward function, and ties are broken arbitrarily. Given a space of environment parameterizations P , the *contextual bandits* problem is to design a space of belief parameterizations B , and to determine the optimal belief-update function $f^* := \arg\max_{f \in \Delta(B)^{B \times X \times A \times \mathbb{R}}} \sum_t \mathbb{E}_{f, \pi_{\beta_t}, \mathcal{R}, \mathcal{T}}[r_t]$.

Now, suppose that an agent follows a bandit-type policy for T time steps; this would generate an *observational dataset* of contexts and actions $\mathcal{D} := \{x_{1:T}, a_{1:T}\}$ (adhering to the contextual bandits literature, we refer to states as “contexts” hereafter). But since rewards r_t and beliefs β_t are quantities *internal* to the decision-maker, we assume that they are not

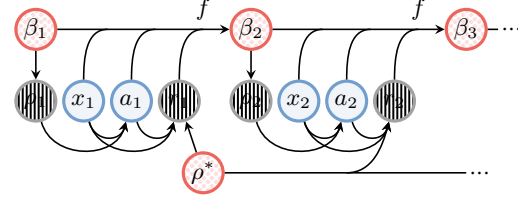


Figure 2. *Graphical Model for ICB.* We aim to infer the quantities in red (dotted) give the quantities in blue (solid) ones.

part of the observational dataset: The former represents the agent’s preferences over outcomes after observing contexts and taking actions, which is not explicitly observable; likewise, the latter represents the agent’s beliefs about what kinds of outcomes their actions result in, which is also not explicitly observable. Thus we ask the novel question:

From the dataset $\mathcal{D}$, can we infer the true environment parameter ρ_{env} , as well as the belief trajectory $\beta_{1:T}$?

Definition 2 (Inverse Contextual Bandits) Consider again a contextual bandit problem $\mathbb{D} := (X, A, \mathcal{R}, \mathcal{T})$, and recall that dynamics $\mathcal{R}, \mathcal{T}$ are unknown to the agent. Given an observational dataset $\mathcal{D}$, and parameter families P, B for environments and beliefs, the *inverse contextual bandits* problem (“ICB”) is to determine the true environment parameter ρ_{env} and the belief parameters $\beta_{1:T}$ (see Figure 2).

It is important to observe that the novelty here is *general*—that is, in posing the “inverse” question of how knowledge evolves over time. Focusing our attention on contextual bandits simply represents a specific choice of problem setting: In this first work on tackling the *inverse* problem in modeling non-stationary agents, it presents a more tractable problem for analysis, and is moreover especially suited for our motivating application in organ allocations. Note that in the bandit setting, the fact that transition dynamics are unknown to the agent is ultimately inconsequential due to greedy policies; we refer to ρ simply as “reward parameter” hereafter. We leave generalization to arbitrary transition dynamics $\mathcal{T} \in \Delta(X)^{X \times A}$ to future work. Also note that ICB itself is not a bandit problem (see Appendix C).

Remark 1 (Environment vs. State Beliefs) When speaking of inferring “beliefs” here in our setting, we are referring to *environment beliefs*—that is, an agent’s knowledge of the environment’s rewards. It is important to distinguish this from *state beliefs* in partially-observed environments [33, 34]—that is, an agent’s estimate of which latent state they are in. State beliefs are computed by an agent using their (known) environment parameters, whereas environment beliefs concern the environment’s (unknown) parameters themselves. State beliefs can be easily inferred [35]: all factors that contribute to state-belief updates (i.e., observations and actions) are observable; on the other hand, environment beliefs are more technically challenging due to latent factors (i.e., internal rewards $r_{1:T}$) that are never observable.

Formally, the *forward problem* of contextual bandits can be framed as a POMDP $(S, A, \Omega, \tilde{T}, \mathcal{O}, \tilde{R})$, where $S := X \times P$, $\Omega := X \times \mathbb{R}$ with $\mathbb{R}$ being the reward space, $\tilde{T}(x, \rho, a)[x', \rho'] := \mathcal{T}(x, a)[x']\delta(\rho' - \rho)$ with δ being the Dirac delta function, $\mathcal{O}(x, \rho, a)[x', r] := \delta(x' - x)\mathcal{R}_\rho(x, a)[r]$, and $\tilde{R}(x, \rho, a) := \mathcal{R}_\rho(x, a)$. Then, a forward agent would maintain beliefs $b_t[x, \rho] := \mathcal{P}_{\beta_t}[\rho]$ updated according to some belief-update function f . However, now consider the *inverse problem* of inferring beliefs from observational data. In usual POMDPs (e.g., in [35]), this involves inferring $\{b_t\}$ given a *complete* dataset $\mathcal{D} = \{a_{1:T}, \omega_{1:T}\}$ of actions and emissions (and assuming f is Bayesian). But in the ICB problem, actually $\omega_{1:T} := (x_{1:T}, r_{1:T})$ and we have no access to $r_{1:T}$ at all. Hence, our learning problem involves inferring $\{b_t\}$ given an *incomplete* dataset $\mathcal{D} = \{a_{1:T}, \omega_{1:T} \setminus r_{1:T}\}$ of actions and incomplete emissions (and not necessarily assuming f is Bayesian). As such, tackling ICB with conventional IO-HMM inference methods is not possible either (see Appendix C).

Remark 2 (Subjective vs. Objective Reward) When speaking of learning what “rewards” an agent is optimizing, the novelty here that our objective refers to the full *belief trajectory* $\beta_{1:T}$ of the agent. It is important to distinguish this from simply learning the *ground-truth parameter* ρ_{env} as in typical IRL. The ground-truth parameter is an objective quantity capturing the “prescriptive” notion of what the agent *ought to be* optimizing, whereas the belief trajectory is a subjective quantity capturing the “descriptive” notion of what the agent *appears to be* optimizing. Existing work in learning from non-stationary agents has focused solely on ρ_{env} , which is all that is required for apprenticeship [36, 37]. In contrast, for the goal of understanding behavior (and how it evolves), Definition 2 brings $\beta_{1:T}$ to focus.

3. Bayesian Inverse Contextual Bandits

We operate in the standard contextual bandits setting [26]: Consider a d -dimensional action space A , and suppose at each time t the agent observes a k -dimensional context $x_t[a] \in \mathbb{R}^k$ for each possible action, thus the context space is given by $X := \mathbb{R}^{d \times k}$. Let the space of reward parameters be $P := \mathbb{R}^k$. Rewards are assumed linear with respect to contexts x_t , and normally-distributed with mean $\mathcal{R}_\rho(x, a) = \langle \rho, x[a] \rangle$ and known variance σ^2 ; that is, $\mathcal{R}_\rho(x, a) := \mathcal{N}(\langle \rho, x[a] \rangle, \sigma^2)$ with $\mathcal{N}(\mu, \sigma^2)$ indicating the normal distribution with mean μ and variance σ^2 , and $\langle \cdot, \cdot \rangle$ denoting the inner product. (Note that these assumptions can be relaxed later in Section 3.1 below.)

Belief Updates Before tackling a much more general case, we begin by modeling the agent’s beliefs as Gaussian posteriors over ρ_{env} , and their belief-update function f as Bayesian updates. Formally, beliefs are described by the family of distributions $\mathcal{P}_\beta := \mathcal{N}(\mu, \Sigma)$, where

$\beta := \{\mu, \Sigma\} \in \mathbb{R}^k \times \mathbb{R}^{k \times k}$, and $\mathcal{N}(\mu, \Sigma)$ is the multivariate Gaussian distribution with mean vector μ and covariance matrix Σ . At each time t , given the posterior $\beta_t = \{\mu_t, \Sigma_t\}$ —such that $\rho_{\text{env}}|h_t \sim \mathcal{N}(\mu_t, \Sigma_t)$ —and the observation (x_t, a_t, r_t) , the belief-update $f(\beta_t, x_t, a_t, r_t)$ is the Dirac delta centered at $\beta_{t+1} = \{\mu_{t+1}, \Sigma_{t+1}\}$, with:

$$\begin{aligned} \mu_{t+1} &:= \Sigma_{t+1} \left(\Sigma_t^{-1} \mu_t + \frac{1}{\sigma^2} r_t x_t[a_t] \right) \\ \Sigma_{t+1} &:= \left(\Sigma_t^{-1} + \frac{1}{\sigma^2} x_t[a_t] x_t[a_t]^\top \right)^{-1}. \end{aligned} \quad (1)$$

Here, the initial belief $\beta_1 = \{\mu_1, \Sigma_1\}$ represents the agent’s (unknown) prior over ρ_{env} .

Finally, to facilitate Bayesian analysis, we consider soft-optimal policies as usual (see e.g., [38]). Formally, instead of picking actions uniformly from $\arg\max_{a \in A} \mathcal{R}_{\rho_t}(x_t, a)$, they are chosen as follows:

$$\pi_{\mathcal{R}_{\rho_t}}^*(x_t)[a_t] := e^{\alpha \mathcal{R}_{\rho_t}(x_t, a_t)} / \sum_{a \in A} e^{\alpha \mathcal{R}_{\rho_t}(x_t, a)}, \quad (2)$$

where $\alpha \in \mathbb{R}_+$ adjusts the stochasticity of action selection ($\alpha \rightarrow \infty$ recovers the hard-optimal case).

Bayesian Learning We propose an expectation-maximization (EM)-like algorithm that estimates the true reward parameter ρ_{env} and the initial belief parameter β_1 , and samples the belief trajectory $\beta_{2:T}$ in alternating steps. First, the joint distribution of all quantities of interest is:

$$\begin{aligned} \mathbb{P}(\rho_{\text{env}}, \beta_1, r_{1:T}, \rho_{1:T}, \mathcal{D}) &= \mathbb{P}(\rho_{\text{env}}, \beta_1) \\ &\times \prod_{t=1}^T \underbrace{\mathbb{P}(x_t)}_{\mathcal{T}[x_t]} \underbrace{\mathbb{P}(\rho_t|\beta_t)}_{\mathcal{P}_{\beta_t}[\rho_t]} \underbrace{\mathbb{P}(a_t|x_t, \rho_t)}_{\pi_{\mathcal{R}_{\rho_t}}^*(x_t)[a_t]} \underbrace{\mathbb{P}(r_t|\rho_{\text{env}}, x_t, a_t)}_{\mathcal{R}_{\rho_{\text{env}}}(x_t, a_t)[r_t]}. \end{aligned} \quad (3)$$

Note that since each belief parameter β_t is a deterministic function of the initial belief parameter β_1 and the history h_t (cf. Equation 1), rewards $r_{1:T-1}$ and belief parameters $\beta_{2:T}$ are essentially interchangeable given the initial belief and context-action pairs $\mathcal{D}$ (i.e., one is completely identified by the other). Then, starting with some initial estimates $\hat{\rho}_{\text{env}}$ and $\hat{\beta}_1$ for the true reward parameter and the initial belief parameter, we iteratively obtain better estimates by performing the following two steps:

- *Expectation step:* Compute the expected log-likelihood of $\rho_{\text{env}}, \beta_1$ given current estimates $\hat{\rho}_{\text{env}}, \hat{\beta}_1$:

$$\begin{aligned} \mathcal{Q}(\rho_{\text{env}}, \beta_1; \hat{\rho}_{\text{env}}, \hat{\beta}_1) &:= \\ \mathbb{E}_{r_{1:T}, \rho_{1:T} | \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D}} [\log \mathbb{P}(r_{1:T}, \rho_{1:T}, \mathcal{D} | \rho_{\text{env}}, \beta_1)], \end{aligned} \quad (4)$$

which we shall approximate by sampling reward values and reward parameters $\{r_{1:T}^{(i)}, \rho_{1:T}^{(i)}\}_{i=1}^N$ from the distribution $\mathbb{P}(r_{1:T}, \rho_{1:T} | \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D})$ using Markov chain Monte Carlo (MCMC) methods.

- *Maximization step:* Compute new estimates $\hat{\rho}'_{\text{env}}, \hat{\beta}'_1$ that yield an improved expected log-posterior:

$$\begin{aligned} & \mathcal{Q}(\hat{\rho}'_{\text{env}}, \hat{\beta}'_1; \hat{\rho}_{\text{env}}, \hat{\beta}_1) + \log \mathbb{P}(\hat{\rho}'_{\text{env}}, \hat{\beta}'_1) \\ & > \mathcal{Q}(\hat{\rho}_{\text{env}}, \hat{\beta}_1; \hat{\rho}_{\text{env}}, \hat{\beta}_1) + \log \mathbb{P}(\hat{\rho}_{\text{env}}, \hat{\beta}_1), \end{aligned} \quad (5)$$

which can be achieved using gradient-based methods.

Sampling Rewards and Reward Parameters Observe that reward values $\mathbf{r}_{1:T}$ are Gaussian-distributed when conditioned on reward parameters $\boldsymbol{\rho}_{1:T}$ (which means they can be sampled exactly conditioned on reward parameters):

Lemma 1 Let $\mathbf{r}_{1:T} = [r_1 \cdots r_T] \in \mathbb{R}^T$ and let $X_t = (1/\sigma^2)[x_1[a_1] \cdots x_t[a_t] \mathbf{0}_{k, (T-t)}] \in \mathbb{R}^{k \times T}$, where $\mathbf{0}_{i,j}$ is the zero matrix with dimensions $i \times j$. Then we have that $\mathbf{r}_{1:T} | \boldsymbol{\rho}_{1:T}, \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D} \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \tilde{\Sigma})$, where

$$\begin{aligned} \tilde{\boldsymbol{\mu}} &:= \tilde{\Sigma} \left(X_T^\top \hat{\rho}_{\text{env}} + \sum_{t=2}^T X_{t-1}^\top (\rho_t - \hat{\Sigma}_t \hat{\Sigma}_1^{-1} \hat{\mu}_1) \right) \\ \tilde{\Sigma} &:= \left(\frac{1}{\sigma^2} I + \sum_{t=2}^T X_{t-1}^\top \hat{\Sigma}_t X_{t-1} \right)^{-1}. \end{aligned} \quad (6)$$

Proof. All proofs can be found in Appendix F. $\square$

This motivates a Gibbs-like sampling procedure whereby samples from $\mathbb{P}(\mathbf{r}_{1:T}, \boldsymbol{\rho}_{1:T} | \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D})$ are approximated by sampling $\mathbf{r}_{1:T}$ and $\boldsymbol{\rho}_{1:T}$ in an alternating fashion from their respective conditional distributions $\mathbb{P}(\mathbf{r}_{1:T} | \boldsymbol{\rho}_{1:T}, \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D})$ and $\mathbb{P}(\boldsymbol{\rho}_{1:T} | \mathbf{r}_{1:T}, \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D})$ instead. However, note that sampling reward parameters $\boldsymbol{\rho}_{1:T}$ from its conditional is not as easy to perform exactly as sampling reward values $\mathbf{r}_{1:T}$. We achieve it by first observing that ρ_t 's are independent from each other conditioned on $\mathbf{r}_{1:T}$:

$$\mathbb{P}(\boldsymbol{\rho}_{1:T} | \mathbf{r}_{1:T}, \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D}) \propto \prod_{t=1}^T \mathbb{P}(\rho_t | \hat{\beta}_t) \mathbb{P}(a_t | x_t, \rho_t), \quad (7)$$

Then, we sample each ρ_t in turn by performing a single iteration of the Metropolis-Hastings algorithm using $\mathcal{P}_{\hat{\beta}_t}$ as the proposal distribution. Algorithm 1 (Bayesian ICB) summarizes the overall procedure.

Algorithm 1 Bayesian ICB

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1: Parameters: Parameters  $\sigma^2$  and  $\alpha$ , learning rate  $\eta \in \mathbb{R}_+$ 
2: Input: Dataset  $\mathcal{D}$ , prior  $\mathbb{P}(\rho_{\text{env}}, \beta_1)$ 
3: Initialize  $\hat{\rho}_{\text{env}}, \hat{\beta}_1 \sim \mathbb{P}(\rho_{\text{env}}, \beta_1)$ 
4: loop
5:   Initialize  $\mathbf{r}_{1:T}^{(0)}, \boldsymbol{\rho}_{1:T}^{(0)}$ 
6:   for  $i \in \{1, 2, \dots, N\}$  do
7:     Sample  $\mathbf{r}_{1:T}^{(i)} \sim \mathbb{P}(\mathbf{r}_{1:T} | \boldsymbol{\rho}_{1:T}^{(i-1)}, \hat{\rho}_{\text{env}}, \hat{\beta}_1, \mathcal{D})$  via Lemma 1
8:     for  $t \in \{1, 2, \dots, T\}$  do
9:       Sample  $\rho'_t, \rho''_t \sim \mathcal{P}_{\beta_t}[\hat{\beta}_1, \mathbf{x}_{1:t-1}, \mathbf{a}_{1:t-1}, \mathbf{r}_{1:t-1}^{(i)}]$ 
10:       $p \leftarrow \min\{1, \mathbb{P}(a_t | x_t, \rho'_t) / \mathbb{P}(a_t | x_t, \rho_t = \rho'')\}$ 
11:       $\rho_t^{(i)} \leftarrow \rho'_t$  w.p.  $p$ , and  $\rho''_t$  otherwise
12:    end for
13:  end for
14:   $\hat{\mathcal{Q}}(\rho_{\text{env}}, \beta_1) = \frac{1}{N} \sum_{i=1}^N \log \mathbb{P}(\mathbf{r}_{1:T}^{(i)} | \boldsymbol{\rho}_{1:T}^{(i)}, \mathcal{D} | \rho_{\text{env}}, \beta_1)$ 
15:   $\{\hat{\rho}_{\text{env}}, \hat{\beta}_1\} \leftarrow \{\hat{\rho}_{\text{env}}, \hat{\beta}_1\}$ 
     $+ \eta [\nabla_{\{\rho_{\text{env}}, \beta_1\}} \hat{\mathcal{Q}}(\rho_{\text{env}}, \beta_1)]_{\{\rho_{\text{env}}, \beta_1\} = \{\hat{\rho}_{\text{env}}, \hat{\beta}_1\}}$ 
     $+ \eta [\nabla_{\{\rho_{\text{env}}, \beta_1\}} \log \mathbb{P}(\rho_{\text{env}}, \beta_1)]_{\{\rho_{\text{env}}, \beta_1\} = \{\hat{\rho}_{\text{env}}, \hat{\beta}_1\}}$ 
16: end loop
17: Output:  $\hat{\rho}_{\text{env}}, \{\beta_{1:T}^{(i)}[\hat{\beta}_1, \mathbf{x}_{1:T}, \mathbf{a}_{1:T}, \mathbf{r}_{1:T}^{(i)}]\}_{i=1}^N$ 

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3.1. Nonparametric Bayesian ICB

So far, we have modeled the learning procedure of the agent with Bayesian updates, and estimated the parameters that characterize their learning procedure (i.e., the true reward parameter ρ_{env} and the initial belief parameter β_1). Clearly, this approach can be similarly applied to different parameterizations of the agent's behavior (e.g., using a UCB-based model, or a general function approximator for f). In this section, instead of imposing a particular type of belief-update, we take a nonparametric approach and regard the agent's beliefs $\beta_{1:T}$ more generally as a random process. First, we establish a prior $\mathbb{P}(\beta_{1:T})$ over that belief process, then describe a procedure to sample from the posterior $\mathbb{P}(\beta_{1:T} | \mathcal{D})$.

Gaussian Process Prior Different from above, now let the agent's beliefs be described by the family of distributions $\mathcal{P}_\beta := \mathcal{N}(\beta, \Sigma_P)$ for $\beta \in \mathbb{R}^k$, where the covariance matrix $\Sigma_P \in \mathbb{R}^{k \times k}$ controls the variability of reward parameters sampled from beliefs. Let $\beta_{1:T} = [\beta_1 \cdots \beta_T] \in \mathbb{R}^{k \times T}$; here we consider a multivariate Gaussian process as prior over belief parameters $\beta_{1:T}$:

$$\text{vec}(\beta_{1:T}) \sim \mathcal{N}(\mathbf{0}, \Sigma_T \otimes \Sigma_B), \quad (8)$$

where $\otimes$ denotes the Kronecker product, $\Sigma_T \in \mathbb{R}^{T \times T}$ controls the covariances between different time steps, and $\Sigma_B \in \mathbb{R}^{k \times k}$ controls the covariances between different components of a given belief β_t . Although our methodology is applicable for any arbitrary Σ_T , we shall fix $(\Sigma_T)_{ij} = \min\{i, j\}$ so our prior becomes a multivariate Wiener process, and differences $\delta_t := \beta_t - \beta_{t-1}$ between consecutive beliefs are independent and identically distributed according to $\mathcal{N}(0, \Sigma_B)$ —that is $\mathbb{P}(\delta_t) \propto \exp(-\frac{1}{2} \delta_t^\top \Sigma_B^{-1} \delta_t)$. Intuitively, this means our prior favors belief trajectories that vary smoothly over time, where differences between consecutive beliefs are probabilistically bounded by Σ_B (since larger changes in consecutive beliefs β_t and β_{t+1} are exponentially less likely in relation to Σ_B).

Sampling Beliefs from the Posterior Having established a Gaussian process prior $\mathbb{P}(\beta_{1:T})$, observe that the posterior for $\beta_{1:T}$ is still a Gaussian process when conditioned on the reward parameters $\boldsymbol{\rho}_{1:T}$:

Lemma 2 Let $\boldsymbol{\rho}_{1:T} = [\rho_1 \cdots \rho_T] \in \mathbb{R}^{k \times T}$. Then we have that $\text{vec}(\beta_{1:T}) | \boldsymbol{\rho}_{1:T}, \mathcal{D} \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \tilde{\Sigma})$, with

$$\begin{aligned} \tilde{\boldsymbol{\mu}} &:= \tilde{\Sigma} (I \otimes \Sigma_P)^{-1} \text{vec}(\boldsymbol{\rho}_{1:T}) \\ \tilde{\Sigma} &:= ((\Sigma_T \otimes \Sigma_B)^{-1} + (I \otimes \Sigma_P)^{-1})^{-1}. \end{aligned} \quad (9)$$

Therefore, similar to the parametric version of our algorithm, we can approximate samples from the posterior $\mathbb{P}(\beta_{1:T} | \mathcal{D})$ by sampling $\beta_{1:T}$ and $\boldsymbol{\rho}_{1:T}$ in an alternating fashion from their respective conditional distributions $\mathbb{P}(\beta_{1:T} | \boldsymbol{\rho}_{1:T}, \mathcal{D})$ and $\mathbb{P}(\boldsymbol{\rho}_{1:T} | \beta_{1:T}, \mathcal{D})$, and discarding the samples for $\boldsymbol{\rho}_{1:T}$. Likewise, we sample each ρ_t in turn by performing a single

iteration of the Metropolis-Hastings algorithm. Algorithm 2 (Nonparametric Bayesian ICB) summarizes the overall sampling procedure. Note that unlike Algorithm 1, this nonparametric approach no longer imposes any restrictions whatsoever—e.g., linearity with respect to contexts, normal distribution about the mean—on the form of $\mathcal{R}_\rho$ (see Appendix C for a discussion of differences between Algorithm 1 and 2).

Algorithm 2 Nonparametric Bayesian ICB

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1: Parameters: Covariance matrices  $\Sigma_P$  and  $\Sigma_B$ 
2: Input: Dataset  $\mathcal{D}$ 
3: Initialize  $\rho_{1:T}^{(0)}, \beta_{1:T}^{(0)}$ 
4: for  $i \in \{1, 2, \dots, N\}$  do
5:   Sample  $\beta_{1:T}^{(i)} \sim \mathbb{P}(\beta_{1:T} | \rho_{1:T}^{(i-1)}, \mathcal{D})$  via Lemma 2
6:   for  $t \in \{1, 2, \dots, T\}$  do
7:     Sample  $\rho'_t, \rho''_t \sim \mathcal{P}_{\beta_t^{(i)}}$ 
8:      $p \leftarrow \min\{1, \mathbb{P}(a_t|x_t, \rho_t = \rho') / \mathbb{P}(a_t|x_t, \rho_t = \rho'')\}$ 
9:      $\rho_t^{(i)} \leftarrow \rho'_t$  w.p.  $p$ , and  $\rho''_t$  otherwise
10:  end for
11: end for
12: Output:  $\{\beta_{1:T}^{(i)}\}_{i=1}^N$ 

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4. Related Work

Policy Learning In modeling an agent’s decision-making from their observed behavior, the overarching goal is often in *imitation learning* (i.e., to replicate their actions) or in *apprenticeship learning* (i.e., to match their performance). The former directly parameterizes imitation policies as black-box functions, typically based on behavioral cloning [39, 44, 45] or state-action distribution matching [46, 47, 48]. The latter takes the indirect approach of first inferring some reward function for which the agent’s demonstrations are assumed to be optimal—and on the basis of which an apprentice policy is trained; this is often done via Bayesian [38, 49, 50] or max-ent [51, 52, 53] IRL. In contrast, our overarching goal is in *descriptive modeling* (i.e., to learn interpretable parameterizations for explaining observed behavior, see Appendix C for a detailed discussion on

our interpretability criteria) [8, 9]. For instance, recent work has represented policies in terms of human-understandable rules [10], goals [11], intentions [12], preferences [13], subjective dynamics [14], and counterfactuals [15].

Learning from Variation Precisely, we wish to understand how behavior has changed over time, especially relevant in healthcare as knowledge [23, 24, 25] and practices [16, 17, 18] continuously evolve. While *variation in practice* has been captured in policy learning from multi-modal [19, 20], multi-task [21, 22], and compound-task [41, 54] demonstrations, little has sought to describe *variation over time*. Some research has explored IRL from non-stationary agents, using labeled data generated by agents performing specific policy updates over time [36, 37] or ranked post-hoc by an expert [42, 55]. However, all of these simply attempt to infer the optimal reward (viz. ground-truth ρ_{env}) for the (prescriptive) goal of apprenticeship, revealing nothing about *how* the actual behavior has evolved (viz. trajectories $\beta_{1:T}$) for the (descriptive) goal of understanding. The only close attempt has simply used change-point detection to allow inferring a sequence of optimal rewards as usual [43].

Bandits Setting To the best of our knowledge, this is the first formal attempt at learning interpretable representations of non-stationary behavior. In formulating and solving this challenge of learning $\beta_{1:T}$, we have focused on the bandit setting [26, 27, 28]. Now, the general notion of an “inverse” bandit problem had been independently proposed by [56] and [57] as the bandit-based counterpart to apprenticeship from a learning agent [36, 37, 42, 55]. However, they both ignore the presence of contexts (viz. “non-contextual” bandits), and—more importantly—identical to the apprenticeship works above, they only attempt to infer the ground-truth reward ρ_{env} , without regard to the trajectory of evolution itself. Table 1 contextualizes ICB with respect to related works in learning from decision-making behavior.

Table 1. *Comparison with Related Work.* We aim to provide an [1] *interpretable* account (i.e. reward-based or white-box) of [2] *non-stationary* behavior (i.e. evolving over time) that explicitly captures the [3] *trajectory* of changes itself (i.e. the agent’s knowledge) in a [4] *stepwise* fashion (i.e. versus batched intervals), and [5] *shares information* between consecutive estimates (i.e. β_t, β_{t+1} are not independent).

	Problem Formulation	Axis of Variation	Learning Target	Interpretable Output ¹	Non-Stationary Agent ²	Trajectory of Changes ³	Stepwise Evolution ⁴	Information Sharing ⁵	Examples
Policy Learning	Imitation Learning	-	π_b	✗	✗	-	-	-	[39]
	Apprenticeship Learning	-	ρ^*	✓	✗	-	-	-	[40]
	Interpretable Policy Learning	-	π_b	✓	✗	-	-	-	[14]
Variation in Practice	Multi-Modal Imitation	K clusters	$\{\pi_{b,k}\}$	✗	✗	-	-	-	[20]
	Compound-Task Learning	K sub-tasks	$\{\pi_k^*\}$	✗	✗	-	-	-	[41]
	Multi-Task Apprenticeship	K clusters	$\{\rho_k^*\}$	✓	✗	-	-	-	[22]
Variation over Time	Ranked Reward Extrapolation	N samples	ρ^*	✓	✓	✗	✗	✗	[42]
	Inverse Soft Policy Improvement	M intervals	ρ^*	✓	✓	✗	✗	✗	[36]
	Change-Points Reward Learning	M intervals	$\{\rho_m^*\}$	✓	✓	✓	✗	✗	[43]
	Inverse Contextual Bandits	T time steps	$\rho^*, \{\beta_t\}$	✓	✓	✓	✓	✓	(Ours)

5. Illustrative Examples

Three aspects of our approach deserve empirical demonstration, and we shall highlight them in turn:

- *Explainability*: ICB should help us understand how medical practice has changed over the years, which is our primary motivation in developing ICB.
- *Belief Accuracy*: ICB should recover accurate beliefs in a robust manner across a variety of learning agents without being sensitive to the underlying learning procedure.
- *Reward Accuracy*: ICB should recover accurate ground-truth reward parameters in a similarly robust manner and “extrapolate beyond” suboptimal demonstrations.

Decision Environments We consider data from the Organ Procurement & Transplantation Network (“OPTN”) as of Dec. 4, 2020, which consists of patients registered for liver transplantation from 1995 to 2020 [58]. We are interested in the decision-making problem of matching organs that become available with patients who are waiting for a transplantation. For each decision, the action space A_t consists of patients who were in the waitlist at the time of an organ’s arrival t , while the context $x_t[a]$ for each patient $a \in A_t$ includes the features of both the organ and the patient. We consider $k = 8$ features: {ABO mismatch, age, creatinine, dialysis, INR, life support, bilirubin, weight difference}. Appendix A discusses our feedback model in more detail.

In addition to the OPTN dataset, to better validate the performance of our method in a more controlled manner, we also devise a semi-synthetic decision environment, where the features $\{x_t[a]\}_{a \in A_t}$ of potential organ-patient pairs are taken from the OPTN dataset, but the organs are matched with a final patient $a_t \in A_t$ for transplantation according to the policies of a variety of simulated learning agents. We determine the ground-truth reward parameter ρ_{env} of this semi-synthetic environment by performing ordinary linear regression over the survival time of patients who actually underwent a transplantation.

Learning Agents For the semi-synthetic experiments, we generate observational datasets by employing the following simulated agents, which includes a stationary agent as well as six (non-stationary) learning agents:

- *Stationary*: The agent knows the ground-truth ρ_{env} and takes actions accordingly (i.e., $\rho_t = \rho_{\text{env}}$ for all t).
- *Sampling*: The agent employs the posterior sampling-based bandit strategy proposed in [26] (cf. Equation 1).
- *Optimistic*: The agent selects actions based on optimistic estimates of their rewards; this is LinUCB in [27].
- *Greedy*: The agent only exploits their knowledge greedily and does not explore their environment effectively.
- *Stepping*: The agent first explores the environment with uniform preferences until some time t^* , whence they learn the

ground-truth reward parameter ρ_{env} and immediately begin taking actions accordingly. In healthcare, stepping behavior like this may occur when new guidelines are introduced.

- *Linear*: The agent learns the true reward parameter ρ_{env} in a linear fashion, starting with uniform preferences.
- *Regressing*: The agent first acquires the ground-truth reward parameter ρ_{env} gradually until some time t^* , at which point they begin to regress while retaining some amount of knowledge. This is a particularly challenging setting because the regressing agent’s behavior does not improve monotonically.

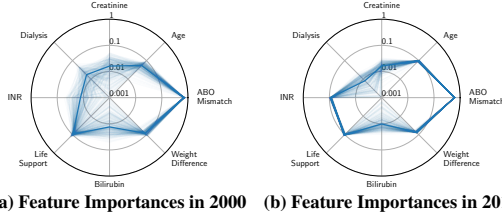
Benchmark Algorithms We consider all the applicable algorithms in the literature as well as our algorithms Bayesian ICB (**B-ICB**) and Nonparametric Bayesian ICB (**NB-ICB**):

- **B-IRL** [38]: This is the classic Bayesian inverse reinforcement learning algorithm that has been widely applied to a variety of problem settings [49, 59, 60]. As usual, it assumes $\rho_t = \rho_{\text{env}}$ for all $t \in \{1, \dots, T\}$.
- **M-fold-IRL**: This runs a copy of Bayesian IRL for each of M equal-sized intervals in the dataset. Note that setting $M = T$ is equivalent to assuming that beliefs over time are completely independent from each other.
- **CP-IRL** [43]: This recently-proposed method runs M copies of Bayesian IRL as well but it employs a change-point detection algorithm to learn best places to divide the dataset (hence referred to as “CP”-IRL).
- **I-SPI** [36]: This recently-proposed method assumes the agent updates their behavioral policy via “soft policy improvements”, which can be “inverted” to recover ρ_{env} (hence referred to as inverse “SPI”). In our setting, this is equivalent to assuming the agent is less stochastic over time (i.e., larger values of α). However, I-SPI only recovers the ground-truth ρ_{env} , and does not provide any estimate for belief trajectories.
- **T-REX** [42]: This recently-proposed method learns ρ_{env} using rankings between context-action pairs; absent explicit rankings, it assumes that pairs encountered later in time are preferred over earlier ones (i.e., $(x_t, a_t) \prec (x_{t'}, a_{t'})$ if $t < t'$). Like I-SPI, T-REX only infers ρ_{env} , not beliefs.

In addition to these benchmark algorithms, as a baseline we also report the performance of simply estimating all preferences to be uniform—that is, $\hat{\rho}_{\text{env}} = \hat{\rho}_t = -\mathbf{1}/k$ (**Baseline**). Details regarding both the learning agents and the benchmark algorithms can be found in Appendix D.¹

Explainability First, we direct attention to the potential utility of ICB as an *investigative device* for auditing and quantifying behaviors as they evolve. We use NB-ICB to estimate belief parameters $\{\beta_t = \mathbb{E}[\rho_t]\}_{t=1}^T$ for liver transplantations in the OPTN dataset. Since the agent’s rewards are linear combinations of features weighted per their belief parameters, we may naturally interpret the normalized belief parameters $|\beta_t(i)| / \sum_{j=1}^k |\beta_t(j)|$ as the *relative importance* of each feature $i \in \{1, \dots, k\}$. Figure 3 shows the relative importances of all eight features in 2000 and 2010,

¹Code to replicate our main results is made available at <https://github.com/alihanhyk/invconban> and <https://github.com/vanderschaarlab/invconban>.



(a) Feature Importances in 2000 (b) Feature Importances in 2010

Figure 3. Relative Feature Importances in 2000 and 2010. INR gains significant importance—despite being the least important feature initially—with the introduction of MELD in 2002.

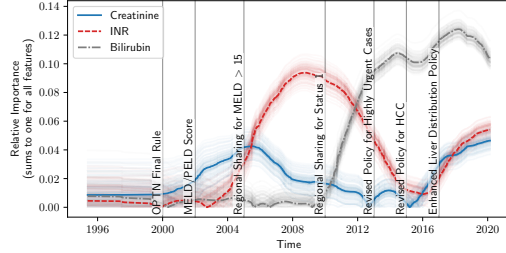


Figure 4. Relative Feature Importances over Time for Creatinine, INR, and Bilirubin. Significant changes in behavior have generally coincided with the important events surrounding guidance on liver allocation policies [62] (note that these were unknown to ICB).

and Figure 4 shows the importance of creatinine, INR, and bilirubin—components considered in the MELD (Model for End-stage Liver Disease) score, a scoring system for assessing the severity of chronic liver disease [61]. Empirically, three observations immediately stand out: First, INR and creatinine appear to have gained significant importance over the 2000s, despite being the least important features in 2000. Second, their importances appear to have subsequently decreased towards the end of the decade. Third, since 2015 their importances appear to have steadily increased again.

Interestingly, we can actually verify that these findings are perfectly consistent with the medical environments of their respective time periods. First, the MELD *scoring system* itself was introduced in 2002, which—using INR and creatinine as their most heavily weighted components—explains the rise in importance of those features in the 2000s. More specifically, not only these factors are weighted positively as in MELD but also their weights evolve in a direction that is consistent with the introduction of MELD (i.e., they are weighted *more and more* positively over time). Second, over time there was an increase in the usage of MELD *exception points* (i.e., patients getting prioritized for special conditions like hepatocellular carcinoma, which are not directly reflected in their laboratory MELD scores), which explains the decrease in relative importance for such MELD components. Third, 2015 saw the introduction of an *official cap* on the use of MELD exception points (e.g., limited at 34 for hepatocellular carcinoma), which is consistent with the subsequent increase in relative importance of those features.

Figure 4 also plots important historical events that happened regarding liver allocation policies [62]. Of course, ICB has

Table 2. Mean Error of Belief Estimates. B-ICB and NB-ICB are best across all types of agents. Note that I-SPI and T-REX are not applicable as they do not estimate belief trajectories. Standard error for is given in parentheses (for five runs).

Algorithm	Learning Agent						
	Stationary	Sampling	Optimistic	Greedy	Stepping	Linear	Regressing
Baseline	0.477 (.00)	0.395 (.03)	0.391 (.03)	0.385 (.02)	0.238 (.00)	0.238 (.00)	0.238 (.00)
B-IRL	0.252 (.24)	0.206 (.09)	0.241 (.19)	0.248 (.20)	0.262 (.02)	0.165 (.01)	0.154 (.01)
5-fold IRL	0.294 (.04)	0.369 (.07)	0.313 (.04)	0.310 (.03)	0.333 (.04)	0.289 (.03)	0.292 (.07)
10-fold IRL	0.424 (.01)	0.407 (.06)	0.401 (.06)	0.399 (.05)	0.398 (.02)	0.386 (.01)	0.388 (.04)
T-fold IRL	1.168 (.01)	1.134 (.02)	1.140 (.01)	1.135 (.01)	1.117 (.01)	1.103 (.01)	1.105 (.01)
5-fold CP-IRL	0.272 (.04)	0.334 (.03)	0.280 (.02)	0.277 (.02)	0.294 (.04)	0.282 (.02)	0.278 (.01)
10-fold CP-IRL	0.409 (.03)	0.462 (.07)	0.388 (.05)	0.384 (.05)	0.374 (.04)	0.376 (.04)	0.401 (.03)
B-ICB	0.120 (.03)	0.140 (.01)	0.121 (.01)	0.120 (.01)	0.234 (.01)	0.153 (.01)	0.147 (.02)
NB-ICB	0.201 (.02)	0.178 (.01)	0.152 (.04)	0.149 (.03)	0.150 (.01)	0.134 (.01)	0.140 (.01)

Table 3. Mean Error of Ground-Truth Reward Estimates. B-ICB is best across all types of agents. M -fold IRL, CP-IRL, and NB-ICB are not applicable as they do not estimate ρ_{env} . Standard error is given in parentheses (for five runs).

Algorithm	Learning Agent						
	Stationary	Sampling	Optimistic	Greedy	Stepping	Linear	Regressing
Baseline	0.477 (.00)	0.477 (.00)	0.477 (.00)	0.477 (.00)	0.477 (.00)	0.477 (.00)	0.477 (.00)
B-IRL	0.252 (.24)	0.224 (.11)	0.258 (.16)	0.266 (.18)	0.237 (.03)	0.266 (.02)	0.251 (.02)
I-SPI	0.231 (.21)	0.206 (.09)	0.276 (.20)	0.263 (.18)	0.237 (.03)	0.266 (.02)	0.250 (.02)
T-REX	1.482 (.06)	1.463 (.06)	1.480 (.05)	1.479 (.06)	1.447 (.08)	1.428 (.08)	1.438 (.98)
B-ICB	0.121 (.03)	0.149 (.05)	0.148 (.03)	0.141 (.04)	0.161 (.02)	0.225 (.02)	0.246 (.02)

no knowledge of these events during training, so any apparent changes in behavior in the figure are discovered solely on the basis of organ-patient matching data in the OPTN dataset. Intriguingly, the importance of bilirubin appears to have not increased until 2008, instead of earlier when the MELD score was first introduced. Now, there are possible clinical explanations for this: For instance, bilirubin is not weighted as heavily as other features when computing MELD scores, so their importance may not have been apparent until the later years, when patients generally became much sicker (with higher MELD scores overall). In any case, however, the point here is precisely that ICB is an *investigative device* that allows introspectively describing how policies have changed in this manner—such that notable phenomena may be duly investigated with a data-driven starting point (see Appendix C for a discussion on how to interpret behavior with ICB).

Belief Accuracy Since it is not possible to compare the belief parameters of different algorithms directly, we define $\|\mathbb{E}_{\rho \sim \mathcal{P}_{\beta_t}}[\rho] - \mathbb{E}_{\rho \sim \mathcal{P}_{\hat{\beta}_t}}[\rho]\|_1$ as the error of belief estimate $\hat{\beta}_t$ at time t . Then, Table 2 shows the mean error of belief estimates learned by various algorithms in our semi-synthetic environment. As we would expect, B-ICB performs the best for the Sampling, Optimistic, and Greedy agents—which learn via Bayesian updates—while NB-ICB—which is more flexible in terms how it models belief trajectories—performs the best for the Stepping, Linear, and Regressing agents. Interestingly, we also observe that both M -fold IRL and CP-IRL perform worse than vanilla IRL. This could be due to the fact that—in both algorithms—the estimates for each interval are trained with fewer data points and independently from the other intervals; that is, there is no *information sharing* and potential similarities between adjacent intervals

are disregarded. In contrast, both B-ICB and NB-ICB formalize some relationship between beliefs at different time steps (viz. Equations 1 and 8); under either formulation, beliefs at different time steps are never independent from each other. Finally, it is worth noting that NB-ICB is capable of capturing the sudden change in the Stepping agent’s behavior despite assuming “smoothly” evolving beliefs (see Appendix A for an additional example). More experiments that evaluate belief accuracy can be found in Appendix E.

Reward Accuracy Defining $\|\rho_{\text{env}} - \hat{\rho}_{\text{env}}\|_1$ as the error in estimating the ground-truth parameter ρ_{env} , Table 3 shows the mean error for various algorithms. B-ICB performs the best; this shows that not only can it recover (subjective) descriptors of how the agent *appears to be* behaving over time, but can also infer the (objective) prescriptor of how the agent *ought to be* behaving ideally. That is, B-ICB can also “extrapolate beyond” suboptimal demonstrations to infer the ground-truth reward. Interestingly, T-REX completely fails: This is because in the contextual bandits setting, it effectively assumes later context-action pairs (i.e., generated by a better policy) on average earn larger rewards than earlier ones (i.e., generated by a worse policy)—but this ignores the stochastic arrival of contexts themselves, which (in contextual bandits) is independent of the policy! For instance, sicker patients at a later time may always yield worse outcomes, no matter how much better the allocation policy has become. Appendix A discusses in much more detail why existing methods fail in the ICB setting.

6. Conclusion

We motivated the importance of learning interpretable representations of non-stationary behavior, formalized the problem of inferring belief trajectories from data, and proposed concrete algorithms with demonstrated advantages in explainability and accuracy over existing methods. Two points deserve brief comment in closing. First, we focused on contextual bandits; while this encapsulates a large class of applications, future work may investigate generalizing this approach to environments with arbitrary transition dynamics. Second, it is crucial to keep in mind that ICB does not claim to identify the real intentions of an agent; humans are complex, and rationality is bounded. What it does do, is to provide an interpretable explanation of how an agent is *effectively* behaving, offering a quantitative yardstick by which to investigate hypotheses and understand behavior.

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